

Syllabus and Assessment

EART60071 Measuring and Predicting 2

Syllabus

Academic year: 2026–2027. Semester 1; 15 credits.

Activity	Time and location
Lectures	Fridays, 09:00–11:00; Schuster, Blackett Theatre.
Practicals	Fridays, 12:00–15:00; Nancy Rothwell, 1A 044 (Computer cluster 3).

Programme information

Core module for:

- MPEC;
- MESPOM.

Optional for: N/A.

Instructor

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Assessment

Assessment	Weighting
Formative online assessment - released 15:00, 11 December 2026	0%
Summative online assessment - 2-hour timed Canvas assessment in the examination period	100%

Computing and course material

The module is taught face-to-face and the practicals use hands-on environmental modelling. The *Tools of the Trade* handout explains the course Linux-server workflow, including Windows and macOS access and secure file transfer, together with the optional Docker environment for practising off campus.

Working off campus. EART60071 students have two ways to continue practical work away from campus. If you do not want to install Docker, you can use the University **Parallels** service to log in remotely to an on-campus University computer and then SSH from that computer to the modelling server. This involves **two logins**: first to the University computer through Parallels, and then to the modelling server. Alternatively, you can install the optional EnvModelling Docker environment and run the modelling environment locally on your own computer.

[Module webpage](#). Canvas is the authoritative location for assessment links, server login details, announcements and updated instructions.

Attendance, practice and feedback

Attendance at timetabled teaching and practical activities is expected. Feedback is available through discussion in class, by working through practical results and model output, and through class discussion of issues that arise in the assessments.

The questions in the *Pressure, Temperature, Humidity, Visibility and Particulate* and *Dynamical Models* chapters are intended as practice. Work through them yourself before looking at the worked answers; use the answers to identify anything you need to revisit or ask about in class.

Assessment details

The module has one summative online assessment worth 100% of the module mark. It is a timed, 2-hour assessment completed on Canvas under examination conditions during the examination period. A formative assessment is provided beforehand so that you can practise the same style of questions and identify topics to revisit before the summative assessment.

Formative online assessment - practice (0%)

Released at 15:00 on 11 December 2026 on Canvas. It has the **same format as the final assessment: 10 questions**. It does not contribute to your module mark. The questions are **randomised**, so you can complete the assessment several times to gain more practice.

You will receive feedback after completing an attempt. Use this to identify topics to revisit, then return to the relevant lecture notes, chapter questions and practical work before trying another randomised attempt.

There will also be time to discuss the formative assessment and get feedback in class before Christmas. After completing an attempt, bring questions about anything you did not understand so that we can work through the reasoning together. Any updated instructions will be given on Canvas.

Summative online assessment - 100%

Timed assessment: 2 hours, on Canvas, during the examination period. The assessment contains **10 questions** and is completed **under examination conditions**. It follows the same format as the formative assessment. The assessment is **open book**: you may bring and consult your own notes during the assessment. The exact date, time and examination arrangements will be confirmed through the University's examination timetable and Canvas.

Once the assessment begins, you have 2 hours to complete it. You should therefore prepare your notes in advance so that useful material is easy to find during the assessment. Questions may draw on material from across the module, including the models used, the assumptions behind them, and the interpretation of practical results. Preparation should involve doing calculations and explaining reasoning, rather than relying on searching through notes during the assessment.

How to prepare effectively

- Work through the questions in the *Pressure, Temperature, Humidity, Visibility and Particulate* and *Dynamical Models* chapters without looking at the worked answers first. Then check your method as well as your final number.
- Revisit practical exercises and make sure you understand what each model calculates, the important inputs and assumptions, and what the output means physically.
- Use the randomised formative assessment several times for additional practice. Review the feedback and revise weaker topics between attempts.
- If you can get as far as explaining *why* an answer is right, rather than only recognising the correct option, you are preparing at the right level.